

Vehicle Routing and MIP

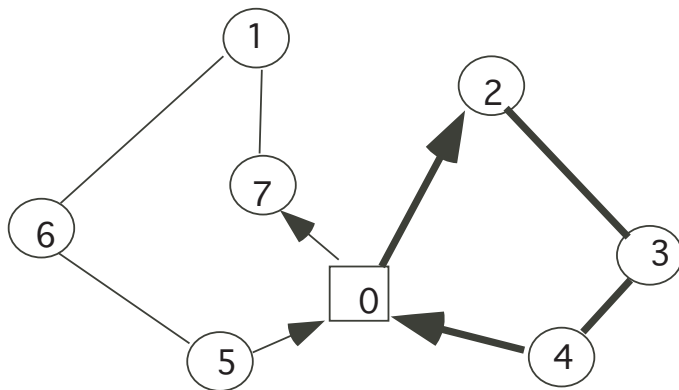
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- The Capacitated Vehicle Routing Problem
- Subproblems: Trees and the TSP
- CVRP
 - Cutting Planes
 - Column Generation
 - Robust Branch-and-Price-and Cut
- An Inventory Routing Problem

The Capacitated Vehicle Routing Problem



- Single Depot. Single Period
- K vehicles can make tours beginning and ending at the depot
- Client i has a demand q_i and is served by exactly one vehicle
- Each vehicle has capacity Q .
- The objective is to find subtours for the K vehicles such that the amount delivered by each vehicle does not exceed Q , the demand of each client is satisfied and the total travel cost/distance is minimized.

Modelling Connectivity: Spanning Trees

Graph $G = (V, E)$ with edge costs c_e for $e \in E$.
Let $y_e = 1$ if e is in the tree.

$$\begin{aligned} \min \quad & \sum_{e \in E} c_e y_e \\ \sum_{e \in E(S)} y_e & \leq |S| - 1 \quad \forall S \subset V \\ \sum_{e \in E} y_e & = |V| - 1 \\ y & \in \mathbb{R}_+^E \end{aligned}$$

where $E(S) = \{e = (i, j) \in E : i, j \in S\}$.

This linear program is tight. The subtour elimination constraints (SECs) describe the convex hull of the spanning trees, but there is an exponential number of constraints.

Separation of SECs

Given a point $\hat{y} \in \mathbb{R}_+^E$, when does it violate one of the SECs?
We formulate this question as an IP.

Let $\alpha_i = 1$ if $i \in S$. We have to check whether there exists $\alpha \in \{0,1\}^V$ such that

$$\sum_{e=(i,j) \in E} \hat{y}_e \alpha_i \alpha_j - \sum_{i \in V} \alpha_i > -1.$$

This is a quadratic 0-1 integer program of the form:

$$\max_{\alpha \in \{0,1\}^V} \alpha^T Q \alpha - p \alpha.$$

It is well known that such IPs can be solved as max flow/min cut problems when $Q \geq 0$.

An Extended “Multicommodity” Formulation

How else can one ensure connectivity?

One way: Construct a graph in which there is a path (one can send a flow) from one node ($r = 1$) to all the others.

f_{ij}^k is the flow (commodity k) in arc (i, j) on the path from r to k .

$$\sum_j f_{ij}^k - \sum_j f_{ji}^k = 0 \quad i \neq k, r$$

$$\sum_j f_{ij}^k - \sum_j f_{ji}^k = 1 \quad i = k$$

$$f_{ij}^k + f_{ji}^k \leq y_e \quad \forall e \in E$$

$$f_{ij}^k \geq 0 \quad \forall i, j, k$$

This formulation is as strong as that with subtours, but it has $O(V^3)$ variables.

The Traveling Salesman Problem: Formulation 1

Symmetric Version $c_{ij} = c_{ji} = c_e$
 $y_e = 1$ if edge e in the tour

$$\begin{aligned} \min \quad & \sum_{e \in E} c_e y_e \\ \sum_{e \in \delta(v)} y_e &= 2 \quad \forall v \in V \\ \sum_{e \in E(S)} y_e &\leq |S| - 1 \quad \forall S \subset V \\ y &\in [0, 1]^E \end{aligned}$$

SECS can be replaced by “cut” inequalities

$$\sum_{e \in \delta(S)} y_e \geq 2 \quad \forall \emptyset \subset S \subset V.$$

The Traveling Salesman Problem: Formulation 2

Weak Extended Formulation (Directed):

u_i is position of node i in the tour starting with node 1 in position 0 ($u_1 = 0$).

$$u_j - u_i \geq x_{ij} - (n-2)(1-x_{ij}) \quad \forall (i,j), j \neq 1$$

$$\sum_j x_{ij} = 1 \quad \forall i$$

$$\sum_i x_{ij} = 1 \quad \forall j$$

$$y_e = x_{ij} + x_{ji} \quad \forall e$$

$$x \in \mathbb{Z}_+^A, u \in \mathbb{Z}_+^n$$

Useful if one wants a compact valid formulation.

Formulations of CVRP

$x_{ij}^k = 1$ if arc (i,j) in tour of vehicle k

$z_i^k = 1$ if client i visited by vehicle k

$$\min \sum_k \sum_{ij} c_{ij}^k x_{ij}^k$$

$$\sum_i x_{ij}^k = z_j^k \quad \forall k, j, j \neq 0,$$

$$\sum_j x_{ij}^k = z_i^k \quad \forall k, i, i \neq 0,$$

$$\sum_j x_{0j}^k = z_0^k \quad \forall k$$

$$\sum_i d_i z_i^k \leq C z_0^k \quad \forall k$$

$$\sum_k z_i^k = 1 \quad \forall i$$

$$x_{ij}^k, z_i^k \in \{0, 1\}$$

Formulations of CVRP with 2 Indices

$$\min \sum_{ij} c_{ij} x_{ij}$$

$$\sum_i x_{ij} = 1 \quad \forall j \neq 0,$$

$$\sum_j x_{ij} = 1 \quad \forall i \neq 0,$$

$$\sum_j x_{0j} = K$$

$$\sum_{ij \in \delta(\bar{S}, S)} x_{ij} \geq \lceil \frac{\sum_{i \in S} d_i}{C} \rceil \quad \forall S \subseteq V$$

$$x_{ij} \in \{0, 1\}$$

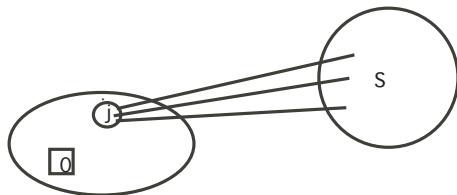
Valid Inequalities and Separation

1. Separation of $\sum_{ij \in \delta(\bar{S}, S)} x_{ij} \geq \lceil \frac{\sum_{i \in S} d_i}{C} \rceil$?

Separation of $\sum_{ij \in \delta(\bar{S}, S)} x_{ij} \geq \frac{\sum_{i \in S} d_i}{C}$ is polynomial.

2. Generalized Large Multistar

$$\sum_{e \in \delta(S)} y_e \geq \frac{2}{C} (d(S) + \sum_{j \notin S} d_j (\sum_{e \in \delta(S: \{j\})} y_e))$$



Column Generation

- A natural idea is to have columns corresponding to feasible subtours for a vehicle.
- The problem is then just to select a set of K subtours such that each client is visited once.
- The trouble with this is that the column generation subproblem is a prize-collecting traveling salesman problem that is almost as hard as the original problem.
- So the typical approach is to look at a larger set of columns that includes all the feasible subtours, but for which the column generation problem is more tractable.

A **q-route** is a walk starting and ending at the depot visiting the clients 0,1, or more times but with total demand at most C .

Note that each time a client i is visited, his demand d_i is counted again.

Suppose $n = 4$, $C = 13$ and $d = (2, 4, 5, 7)$.

A feasible subtour is for example $0 - 3 - 2 - 0$ with $\sum_i d_i = 9$.

A q-route, that is not a feasible subtour, is $0 - 1 - 2 - 1 - 0$ with $\sum_i d_i = 8$.

$0 - 1 - 2 - 3 - 1 - 0$ is a q-route without a 2-cycle as

$\sum_i d_i = 13 \leq C$.

A minimum cost q-route without 2-cycles can be found by dynamic programming.

Naive DP for q-routes

$f(i, j, k)$ is the min cost of a walk starting at the depot and ending with visits to i , then j in which k units are delivered.

$$f(i, j, k) = \min_{p: p \neq j} [f(p, i, k - d_j) + \bar{c}_{ij}]$$

Min Cost q-route is

$$\min_{i, j, k \leq C} [f(i, j, k) + \bar{c}_{0j}].$$

Column Generation with q -routes without 2-cycles

Let q_j^e be the coefficient of edge e in q -route j with variable λ_j .
One has

$$\begin{array}{rcll} \sum_{j=1}^p q_j^e \lambda_j & -x_e & = & 0 \quad e \in E \\ \sum_{j=1}^p \lambda_j & & = & K \\ \sum_{e \in \delta(i)} x_e & & = & 2 \quad i \in V \\ x_e & \geq & & 0 \quad e \in E \\ \lambda_j & \geq & & 0 \quad j = 1, \dots, p \end{array}$$

The Master Problem

$$\begin{array}{rcll}
 \sum_{e \in \delta(i)} x_e & = & 2 & i \in V \\
 \sum_{e \in \delta(0)} x_e & = & 2K & \\
 \sum_{e \in \delta(S)} x_e & \geq & 2k(S) & S \subseteq V \\
 x_e & \leq & 1 & e \in E \setminus \delta(0) \\
 -x_e & = & 0 & e \in E \\
 \sum_{j=1}^p q_j^e \lambda_j & & & \\
 \sum_{j=1}^p \lambda_j & = & K & \\
 x_e & \geq & 0 & e \in E \\
 \lambda_j & \geq & 0 & j = 1, \dots, p
 \end{array}$$

The Master Problem

$$\min \sum_{j=1}^p \sum_{e \in E} l_e q_j^e \lambda_j$$

$$\mu \quad \sum_{j=1}^p \sum_{e \in \delta(i)} q_j^e \lambda_j = 2 \quad i \in V$$

$$\nu \quad \sum_{j=1}^p \sum_{e \in \delta(0)} q_j^e \lambda_j = 2K$$

$$\pi \quad \sum_{j=1}^p \sum_{e \in \delta(S)} q_j^e \lambda_j \geq 2k(S) \quad S \subseteq V$$

$$\omega \quad \sum_{j=1}^p q_j^e \lambda_j \leq 1 \quad e \in E \setminus \delta(0)$$

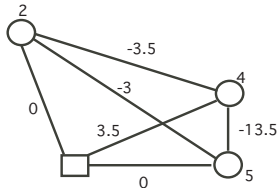
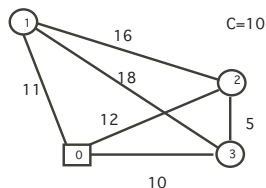
$$\lambda_j \geq 0 \quad j = 1, \dots, p$$

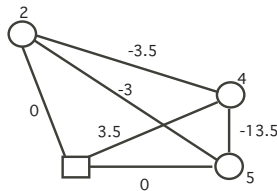
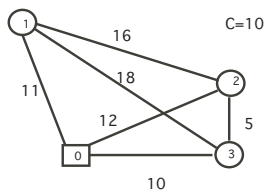
Reduced cost of x_e : $l_e - \mu_i - \mu_j - \sum_{e \in \delta(S)} \pi_S - \omega_e \quad e \in E \setminus \delta(0)$

$l_e - \mu_j - \nu - \sum_{e \in \delta(S)} \pi_S \quad e \in \delta(0)$

$$d = (2, 4, 5), C = 10$$

	w_1	w_2	w_3	w_{12}		u
$cw =$	22	24	20	39	59	
$v = 1$	2	2		2	$=$ 2	11
$v = 2$		2		2	$=$ 2	8.5
$v = 3$			2		$=$ 2	10
$v = 0$	2	2	2	2	$=$ 4	
1, 2	2	2		2	$\geq$ 2	
1, 3	2		2	2	$\geq$ 2	
2, 3		2	2	2	$\geq$ 2	
1, 2, 3	2	2	2	2	$\geq$ 4	
$e = 12$				1	$\leq$ 1	





min cost q-route: 0 – 2 – 3 – 0 with reduced cost of -10

alternative q-route: 0 – 1 – 2 – 1 – 0 with reduced cost of -7

	w_1	w_2	w_3	w_{12}	w_{23}		u
$cw =$	22	24	20	39	27		49
$v = 1$	2	2		2		$= 2$	7.5
$v = 2$		2		2	2	$= 2$	8.5
$v = 3$			2		2	$= 2$	6.5
$v = 0$	2	2	2	2	2	$= 4$	3.5
1, 2	2	2		2	2	≥ 2	
1, 3	2		2	2	2	≥ 2	
2, 3		2	2	2	2	≥ 2	
1, 2, 3	2	2	2	2	2	≥ 4	
$e = 12$				1		≤ 1	
$e = 23$					1	≤ 1	-10

Recent Developments

Recently q -routes replaced by ng -routes

Note that can add constraints $\alpha_e x_e \leq \alpha_0$ and approach still works.

Robust branch-and-cut-and-price.

If cuts on the q -route variables - column generation becomes more difficult

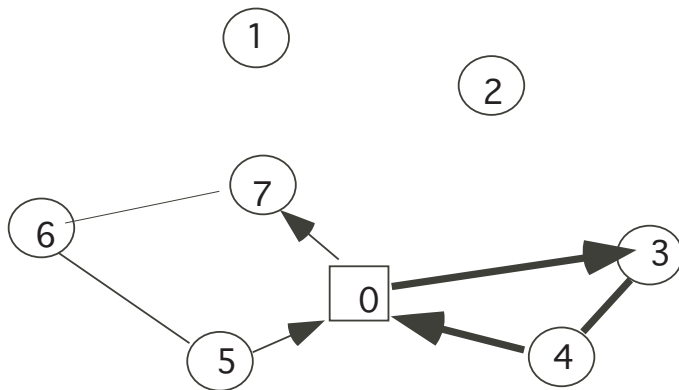
What about **time-windows**?

Similar BCP approach.

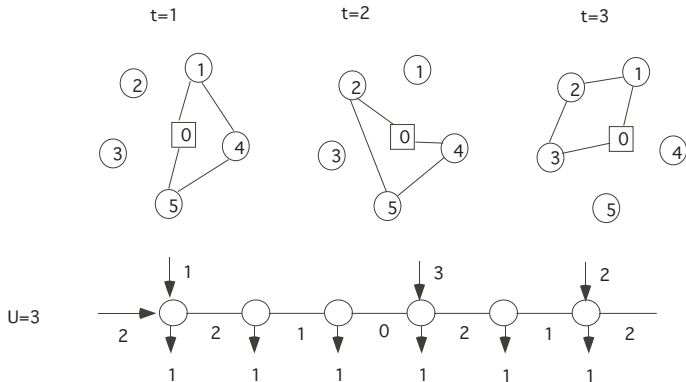
Column generation may be easier - less choice.

Cut generation more difficult - cuts based on infeasibility.

The Inventory Routing Problem: One Period



The Inventory Routing Problem



Model 1: Single Period Inventory Routing Problem

$$\sum_{i \in I} x^i \leq Cz^0, \quad x^i \leq W^i z^i, \quad z^0 \geq z^i, \quad i \in I$$

$$\sum_{j \in I0} y^{ij} = \sum_{j \in I0} y^{ji} = z^i, \quad j \in I0,$$

$$\sum_{i \in S \cup \{0\}} \sum_{j \in I \setminus S} y^{ij} \geq z^i, \quad S \subseteq I,$$

Capacitated Subtours

$$x_i \in R_+^T, z^i \in \{0, 1\} \quad i \in I, z^0 \in \{0, 1, \dots, K\}, y^{ij} \in \{0, 1\}^A$$

First Observations

Where do the capacities come from?

$$W_t^i = \min\{C, d_t^i + \bar{S}^i\}$$

What about the capacitated subtours?

$$C \sum_{i \in I \setminus S} \sum_{j \in S} y^{ij} \geq \sum_{i \in S} x^i$$

where $S \subseteq I$ and $I \setminus 0 = I \cup \{0\}$.

3-index formulation

Direct a flow of x^k from the origin to each client $k \in I$:
 f^{ijk} is the flow in (i,j) with destination $k \in I$

$$\sum_{i \in I_0} f^{ijk} - \sum_{i \in I} f^{jik} = 0 \quad \forall j \neq k, \forall k$$

$$\sum_{i \in I_0} f^{ijk} = x^k \quad k = j, \forall k$$

$$\sum_{k \in I} f^{ijk} \leq C y^{ij} \quad \forall (i,j)$$

$$f^{ijk} \leq W^k y^{ij} \quad \forall i,j,k$$

Metric Inequalities

Suppose $\mu^{ij} \geq 0$ for all $(i,j) \in A$ and $S \subseteq I$, then

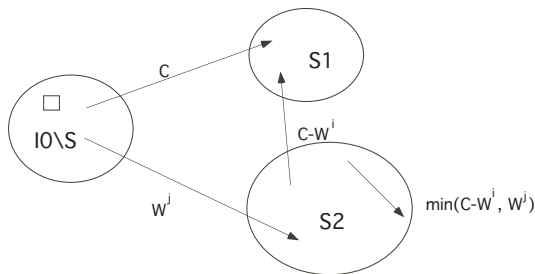
$$\sum_{(i,j) \in A} \mu^{ij} y^{ij} \geq \sum_{i \in S} x^i$$

is a valid inequality if

$$\sum_{(i,j) \in P} \mu^{ij} \geq \min(C, \sum_{i \in v(P) \cap S} w^i)$$

for every subtour P beginning and ending at the depot, where $v(P) \subseteq I$ are the clients on subtour P .

A Particular Metric Inequality



S_1, S_2 is a partition of S

$$\begin{aligned} & \sum_{(IO \setminus S, S_1)} cy^{ij} + \sum_{(IO \setminus S, S_2)} w^j y^{ij} + \sum_{(S_2, S_1)} (c - w^i) y^{ij} \\ & + \sum_{(S_2, S_2)} \min(c - w^i, w^j) y^{ij} \geq \sum_{i \in S} x^i \end{aligned}$$

Example of a Particular Metric Inequality

$$\begin{aligned} & \sum_{(I \setminus S, S_1)} cy^{ij} + \sum_{(I \setminus S, S_2)} w^j y^{ij} + \sum_{(S_2, S_1)} (C - w^i) y^{ij} \\ & + \sum_{(S_2, S_2)} \min(C - w^i, w^j) y^{ij} \geq \sum_{i \in S} x^i \end{aligned}$$

$$|I| = 3, C = 300, W = (289, 123, 76), S = \{1, 2, 3\}, \\ S_1 = \{1\}, S_2 = \{2, 3\}$$

$$\begin{aligned} & 300y^{01} + 123y^{02} + 76y^{03} + (300 - 123)y^{21} + (300 - 76)y^{31} \\ & + \min(76, 300 - 123)y^{23} + \min(123, 300 - 76)y^{32} \geq x^1 + x^2 + x^3 \end{aligned}$$

Multi-Period Metric Inequalities with Stock Upper Bounds: An Example

Add $s_{t-1}^i + x_t^i = D_t^i + s_t^i$ and IRP model for periods $1, \dots, T$.

$$\begin{aligned} C \sum_{u=k}^t \sum_{(i,j) \in (I \setminus S, S)} y_u^{ij} &\geq \sum_{u=k}^t \sum_{i \in S} x_t^i = \sum_{i \in S} \sum_{u=k}^t x_u^i \\ &\geq \sum_{i \in S} (D_{kt}^i - s_{k-1}^i) \end{aligned}$$

With $s_{k-1}^i \leq U^i$, above is of the form:

$$\sigma + Cv \geq b, 0 \leq \sigma \leq h, v \in Z.$$

This gives the inequalities:

$$v \geq \lceil \frac{b-h}{C} \rceil \quad \text{and} \quad \sigma + \rho v \geq \rho \lceil \frac{b}{C} \rceil.$$

Table: VMIRP-SUB: computational results for the instances with $n = 50$ and $T_{max} = 6$

		BC					C&L		
No	LB_{Ini}	LB_{LS}	LB_{Cut}	BUB	Nodes	Time	BLB	BUB	Time
l1	8375	9754	9754	9966	15123	1485	9901	9976	86400
l2	8952	10516	10523	10632	65	334	10632	10632	2536
l3	8725	10376	10391	10511	3972	1876	10511	10511	1355
l4	8628	10243	10243	10513	166667	18016	10513	10513	60289
l5	8386	9860	9899	10113	2500	2327	10113	10113	2416
l6	8417	9945	9948	10148	1900	2318	10114	1014x	86400
l7	8355	9776	9776	9982	284288	28195	9982	9982	14698
l8	8385	10015	10066	10299	878	1360	10253	10229	86400
l9	8484	9897	9904	10010	819	801	10010	10010	6326
l10	8014	9546	9546	9659	2425	2081	9659	9659	3523
h1	29508	29862	29906	30189	1235	645	30189	30189	3036
h2	27983	29601	29615	29790	133	357	29790	29790	3334
h3	27830	29634	29657	29791	219	809	29791	29791	4020
h4	29517	31241	31241	31518	1424	1618	31518	31518	5737
h5	27413	28993	29021	29240	199	565	29240	29240	684
h6	30008	31621	31630	31903	367	1048	31903	31903	28320
h7	27933	29397	29397	29734	7988	1703	29734	29734	13561
h8	23923	25692	25692	25954	328	1202	25954	25954	21552
h9	28467	29863	29884	30193	390	822	30193	30193	20581
h10	29508	31101	31101	31338	83	488	31338	31338	1879

Table: Computational results for the instances 1n15T6 with $n = 15$ and $T_{max} = 6$

		BC					C&L	
	k	Subtour	LS	Prop1	Gen	Cover	BLB	BUB
l1	2	4880.7	5581.1	5681.8	5732.8	5803.6	5987.4	5987.4
l1	3	5640.7	6222.0	6541.4	6580.9	6684.5	6861.07	6861.07
l1	4	6527.8	7049.5	7484.6	7525.6	7622.3	7320.32	7767.75
l1	5	7466.0	7697.3	8377.4	8480.9	8512.7	7574.4	8975.61
h1	2	11388.5	12196.1	12305.6	12367.0	12369.2	12624.7	12624.7
h1	3	12150.2	12820.6	13165.7	13248.6	13310.4	13517.6	13517.6
h1	4	13039.4	13642.5	14111.8	14217.6	14240.1	13999.8	14515.3
h1	5	13975.3	14540.1	15006.9	15109.4	15156.1	14262.2	15470.7

The MIP systems are amazing, but sometimes one can still help.

Thank you for your attention

CVRP



R. Baldacci, N. Christofides, and A. Mingozzi. An exact algorithm for the vehicle routing problem based on the set partitioning formulation with additional cuts. *Mathematical Programming*, 115(2):351–385, 2008.



R. Baldacci, A. Mingozzi, and R. Roberti. New route relaxation and pricing strategies for the vehicle routing problem. *Operations Research*, 59(5):1269–1283, 2011.



R. Fukasawa, H. Longo, J. Lysgaard, M. Poggi de Arag o, M. Reis, E. Uchoa, and R.F. Werneck. Robust branch-and-cut-and-price for the capacitated vehicle routing problem. *Mathematical Programming*, 106(3):491–511, 2006.



Improved Branch-Cut-and-Price for Capacitated Vehicle Routing D. Pecin, A. Pessoa, M. Poggi and E. Uchoa, IPCO 2014 (to appear)

IRP



C. Archetti, L. Bertazzi, G. Laporte, and M.G. Speranza. A branch-and-cut algorithm for a vendor-managed inventory-routing problem. *Transportation Science*, 41(3):382–391, 2007.



P. Avella, M. Boccia, and L.A. Wolsey. Single-item reformulations for a vendor managed inventory routing problem: computational experience with benchmark instances. *CORE Discussion Paper* 2013.



L.C. Coelho and G. Laporte. The exact solution of several classes of inventory-routing problems. *Computers and Operations Research*, 40(2):558 – 565, 2013.